

AVINASH TT

Chennai, Tamil Nadu, India • hadex.code@gmail.com
linkedin.com/in/hadex • github.com/hadexcode-collab

PROFESSIONAL SUMMARY

Results-driven Computer Science & Engineering undergraduate with hands-on experience in full-stack software development, distributed edge computing, artificial intelligence, and autonomous system architecture. Proven ability to design end-to-end scalable applications, optimize backend performance, and integrate complex hardware-software platforms. Adept at leveraging data structures, algorithmic problem solving, and modern development tools to solve complex, real-world problems. Seeking a Software Development Engineer (SDE) or Machine Learning role to leverage expertise in AI, backend performance, and scalable system architecture.

TECHNICAL SKILLS

Programming: Python, Java, C++, SQL, JavaScript

AI/ML & Vision: TensorFlow, PyTorch, OpenCV, Scikit-learn, Whisper AI, Computer Vision

Web & Frameworks: FastAPI, Flask, Vite, Tailwind CSS, REST APIs

Tools & DevOps: Git, GitHub, Docker, Pydantic, Linux/Unix

Systems & Hardware: Microservices, Embedded Systems (ESP32, LoRa SX1278), Autonomous Systems, Edge Computing, Jetson Nano, LiDAR

Spoken Languages: English, Tamil, Hindi, Telugu

PROJECT EXPERIENCE

ASCEND - GPS-Free Autonomous Drone System | ISRO IROC-U 2026

- Architected autonomous perception and state estimation algorithms enabling high-precision navigation and docking in GPS-denied environments without human intervention.
- Integrated real-time computer vision models with embedded flight controllers (Pixhawk V6) and hardware payloads including Jetson Nano and LiDAR, achieving low-latency edge inference for obstacle avoidance.
- Led end-to-end hardware-software system integration and 3D CAD modeling documentation; team successfully qualified for Round 2 in the national ISRO Robotics Challenge.

THADAM / THADAM2 - AI Digital Preservation Platform | Smart India Hackathon

- Engineered an automated archival processing pipeline using Python to ingest, index, and transform unstructured historical multimedia into structured, searchable data formats.
- Integrated Whisper AI and OpenCV models for high-accuracy speech-to-text transcription and video frame analysis, reducing manual data-curation effort.
- Designed a modular, scalable software architecture that streamlined search queries and multimedia data retrieval, optimizing models and migrating legacy logic.

Kalakitchen - Video-to-Recipe AI Assistant

- Engineered a centralized machine learning pipeline to process cooking video inputs, utilizing advanced audio and image recognition techniques.
- Developed robust models to automatically extract complete ingredient lists and step-by-step cooking instructions directly from raw multimedia.

SafeRoute - AI-Assisted Emergency Response Platform | Emergency Response Tech

- Designed a resilient emergency routing system tailored for zero-connectivity and disaster scenarios.
- Implemented long-range mesh communication protocols using LoRa SX1278 and Meshtastic modules on ESP32 microcontrollers to reliably transmit critical location data without cellular networks.
- Developed full-stack data ingestion logic to merge hardware payload signals into a unified tracking dashboard.

CERTIFICATIONS & PROFESSIONAL LEARNING

- **NPTEL:** Software Conceptual Design - Elite Certification (Score: 70/100)
- **Tata ESG Virtual Experience (Forage):** Certificate of Completion (Nov 2025)
- **Deloitte Data Analytics Virtual Experience (Forage):** Certificate of Completion (Jun 2025)
- **Adobe India Hackathon:** Round 1 Online Assessment & Coding Participant (Jul 2025)

ACHIEVEMENTS

- **National Qualifier (Round 2) - ISRO Robotics Challenge (IROC-U 2026):** Recognized for engineering the ASCEND GPS-free autonomous drone system.
- **National Finalist - Smart India Hackathon:** Selected among top national candidates for THADAM, an AI-driven digital preservation solution.

EDUCATION

Sathyabama Institute of Science and Technology, Chennai, India

Bachelor of Engineering (B.E.) in Computer Science and Engineering | CGPA: 7.80/10.0